GRID-CLOCK COSMOLOGY IV

QUANTUM COMPLETION AND EMERGENT GAUGE STRUCTURE

A Fixed-Graph Canonical Quantum-Gravity Candidate with Emergent Gauge Redundancy, Infrared GR Uniqueness, Constraint Closure, Semiclassical Recovery, and Saturated Strong Gravity

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Exploratory companion to the consolidated GCC three-paper series

STATUS NOTE

This manuscript presents an exploratory and effective quantum-gravity completion of Grid-Clock Cosmology (GCC). The v1.1 revision strengthens the origin of the classical gauge and ADM sectors using operational record redundancy, finite-graph label quotients, local-factorization consistency, and a restricted finite-to-continuum refinement proposition. It combines established canonical-gravity and differential-geometric mathematics with GCC-specific reconstruction principles and numerical consistency tests. It does not claim that quantum gravity has been solved, that nature realizes GCC, that the microscopic update is uniquely derived from the original GCC papers, or that the restricted refinement proposition is a general theorem. Results are explicitly separated into established mathematics, numerical reconstructions, low-energy uniqueness statements, effective completion choices, and falsifiable completion-dependent predictions.

Abstract

Grid-Clock Cosmology (GCC) treats a fixed graph of elementary adjacency relations and a universal integer update index as more fundamental than physical distance and proper time. The first three GCC papers formulated the discrete kinematics, an observationally constrained cosmological sector with an exact flat-ΛCDM limit, and a saturation-motivated regular black-hole sector. The present paper develops a canonical quantum-gravity candidate that connects those effective descriptions to a single fixed-graph phase-space and operator framework. The v1.1 revision adds a foundational derivation chain for the infrared gauge structure. The local metric record $h=e\,e^T$ has a three-dimensional kernel identified with local SO(3); smooth address changes are reconstructed as redundancies of relational link records; the exact finite counterpart is a quotient by graph relabelings rather than the automorphism group of a particular geometry; and bounded-strain relabeling sequences converge to smooth pullbacks under refinement. A local-factorization/scheduler criterion then requires overlapping refresh commutators to act only in the record-invisible sector. For normal geometric refreshes this produces the metric structure function $h^{ab}(N\partial_b M-M\partial_b N)$, providing an emergent route to the hypersurface-deformation algebra rather than inserting it as an unexplained continuum axiom. Time-reversal symmetry, low-energy analyticity, local covariance, and the resulting closure reduce the leading gravitational Hamiltonian to the ADM/DeWitt universality class. Independently, metric-compatible torsion-free transport, link curvature, matter universality, primitive triad variables $(e_a{}^i, P^a{}_i)$, Weyl quantization, and finite-regulator constraint tests are constructed explicitly. For the current local Hamiltonian class, whose momentum dependence is at most quadratic and whose curvature term is configuration-only, the Weyl/Moyal expansion gives no intrinsic higher-ħ correction to the H–H commutator; the remaining anomaly tracks the finite-spacing regulator and converges away. A positive master-constraint construction yields a non-empty physical transverse-traceless graviton sector, while a universal-clock history-state prescription permits conditional unitary evolution in K. Coherent physical gravitons recover classical linearized gravitational waves, and scalar, gauge, fermion, and graviton sectors recover a common low-energy relativistic causal cone and standard mode algebras. For the ultraviolet sector we select a bounded local load domain $0\le\chi\le1$ and an explicit spherical regular-core realization $m(r)=Mr^3/(r^3+2M\ell_*^2)$. The center has finite universal curvature, an extremal zero-temperature endpoint occurs at $M_{crit}=(3\sqrt{3}/4)\ell_*$ in geometric units, and the exterior differs from Schwarzschild first at order r^{-4}. The resulting static branch predicts correlated $O((\ell_*/M)^2)$ shifts in the shadow, photon-sphere frequency, damping proxy, and ISCO observables. The construction is therefore a concrete and falsifiable quantum-gravity candidate, while exact microscopic uniqueness, a general graph-refinement theorem, rotating strong-gravity solutions, nonlinear physical-spectrum control, and a dynamical evaporation/Page-curve derivation remain open.

Keywords: quantum gravity; emergent spacetime; discrete substrate; canonical gravity; constraint algebra; Weyl quantization; master constraint; regular black holes; falsifiability

1 Introduction

General relativity is extraordinarily successful as a classical theory of spacetime, but its canonical quantization remains difficult because the dynamical geometry is constrained, the Hamiltonian constraint algebra closes with structure functions, and regulator or factor-ordering choices can generate anomalies [4,5]. Discrete approaches add a second requirement: the microscopic theory must possess a well-defined local update while reproducing local Lorentz symmetry, general covariance, and ordinary quantum field theory at long wavelengths. GCC was formulated to explore this hierarchy from a fixed graph and a universal update ordering [1–3].

Papers I–III deliberately stopped short of a full quantum-gravity completion. Paper I defined the fixed graph, universal clock K, effective lapse/shift/spatial metric, and local unitary update as foundational or correspondence-level structures [1]. Paper II showed that the minimum homogeneous cosmological sector is observationally driven close to its exact n=3 flat-ΛCDM limit [2]. Paper III studied a Hayward-type saturated strong-gravity ansatz and emphasized that curvature regularity alone is not a microscopic quantum theory of black holes [3,13]. The present paper addresses the missing canonical and quantum layers.

The strategy is constructive rather than axiomatic. We first ask which geometric and gauge structures can be understood as redundancies of a fixed-graph relational representation rather than assumed from continuum general relativity. The local triad-to-metric map is used to reconstruct internal frame redundancy; finite label quotients and smooth refinement sequences are used to recover spatial address gauge; and local-factorization consistency is used to test whether normal refresh commutators are forced into that gauge sector. We then ask whether the remaining low-energy Hamiltonian freedom collapses to the ADM/DeWitt form. Matter sectors test universality of the emergent metric. Only after these classical gates are passed are primitive variables quantized, the operator ordering fixed, the quantum constraint algebra audited, and a physical Hilbert-space prescription constructed. Finally, a bounded strong-gravity completion is selected and translated into explicit observational predictions.

This paper is intentionally conservative about status. Several mathematical ingredients—Levi-Civita uniqueness, ADM dynamics, the DeWitt supermetric, hypersurface-deformation uniqueness results, Weyl/Moyal quantization, master-constraint constructions, and regular black-hole metrics—are established in the literature [4,5,10–13,19]. The quotient GL^+(3)/SO(3), tensor pullback laws, and Lie derivatives used in the record analysis are standard differential geometry. The GCC-specific content is the fixed-graph interpretation, the operational record hierarchy, the numerical scheduler/refinement tests, the primitive refresh completion, the chosen clock/history implementation, the bounded-load ultraviolet rule, and the resulting one-scale prediction set. These distinctions are maintained throughout.

2 Relation to the GCC series and completion hierarchy

2.1 Fixed graph and universal clock

The microscopic spatial substrate is a graph $\mathcal{G}=(\mathcal{V},\mathcal{E})$ with a fixed set of adjacency relations and constant baseline vertex count. The universal update index $K\in\mathbb{Z}$ orders microscopic refreshes. Physical proper time and spatial distance are emergent coarse-grained quantities. In the reference vacuum one update and one elementary link may be calibrated to t_P and ℓ_P, while low-energy local Lorentz invariance is a mandatory correspondence condition [1].

$$ds^2 = -\alpha^2 c^2 t_P^2 dK^2 + \ell_P^2 h_{ab} (dN^a + \beta^a dK)(dN^b + \beta^b dK) \tag{2.1}$$

Here α, β^a, and h_{ab} are emergent lapse, shift, and spatial metric fields. The fixed graph is not interpreted as literally stretching. Instead, h_{ab} changes the effective physical distance associated with graph separations. This distinction is essential: a lapse-only model cannot reproduce the full tensorial content of general relativity, including the correct null and lensing structure.

2.2 Completion hierarchy

Table 1. Logical hierarchy used in the present completion.

Level	Content	Status
Foundational	Fixed graph, universal update K, local reversible/unitary microscopic evolution	Original GCC architecture
Emergent geometry	Metric-compatible torsion-free transport; operational record gauge; smooth label-quotient/refinement sector; link curvature; ADM/DeWitt infrared universality	Reconstructed / tested; restricted refinement proposition

Level	Content	Status
Matter universality	Scalar, U(1), SU(2), and spinor sectors couple to the same emergent metric	Numerically tested at low energy
Canonical quantum layer	Primitive triad phase space, Weyl quantization, constraint closure	Exploratory completion
Physical states	Master constraint plus universal-K history states	Exploratory completion
UV completion	Bounded load domain and regular-core branch	Selected effective ansatz
Predictions	Strong-field one-scale deviations and critical endpoint	Falsifiable consequences of selected branch

3 Fixed-graph reconstruction of spatial geometry and the GR kinetic structure

3.1 Information norm and infinitesimal address consistency

At each graph cell let h_{ab} be interpreted as the local inner product for coordinate components of physical information vectors. Norm-preserving transport across a link from cell i to j is represented by U satisfying $U^T h_j U=h_i$. Expanding $U=I-a\Gamma+O(a^2)$ gives the metric-compatibility condition. A second principle requires two infinitesimal orthogonal refreshes to have no spurious $O(a^2)$ translation; in the continuum this is the torsion-free condition. Together these conditions select the Levi-Civita connection. Numerically, a centered infinitesimal link implementation recovered the continuum connection with second-order convergence and reconstructed curvature with the same order.

$$D_k h_{ab} = \Gamma^c_{ak} h_{cb} + \Gamma^c_{bk} h_{ac}, \quad \Gamma^a_{bk}=\Gamma^a_{kb} \tag{3.1}$$

The key interpretation is not that Levi-Civita geometry is new mathematics, but that a fixed graph can carry the same geometry through norm-preserving link transport and infinitesimal address consistency. Plaquette holonomy then reproduces Riemann curvature, while metric matching alone is insufficient: a control link that exactly preserves endpoint norms but does not encode torsion-free connection structure fails to reconstruct curvature.

3.2 End-to-end curvature reconstruction and DeWitt selection

The reconstructed connection was inserted into a differentiable curvature pipeline rather than using an analytic Ricci formula inside the Hamiltonian. For a diagonal but geometrically three-dimensional metric, the derived scalar curvature error converged as $a^{5.43}$ under a sixth-order finite-difference regulator. The canonical kinetic family was parameterized by λ,

$$\mathcal{H}_\lambda = (\pi^{ab} \pi_{ab} - \lambda \pi^2)/\sqrt{h} - \sqrt{h}\, R_{\text{derived}} \tag{3.2}$$

and compared with the hypersurface-deformation target generated by the spatial diffeomorphism constraint. The signed closure root approached the general-relativistic DeWitt value $\lambda=1/2$: at $N=33$ the root was 0.500045, the distance from 1/2 was 4.5×10^{-5}, and the closure anomaly at $\lambda=1/2$ was 1.04×10^{-4}. The curvature error, closure anomaly, and root offset all converged with powers near $a^{5.5}$, supporting a common regulator-error origin rather than an alternative continuum kinetic coefficient.

Table 2. End-to-end derived-curvature and DeWitt-selection audit (N=33 finest grid).

Diagnostic	Finest value	Convergence
Derived scalar curvature	6.5×10^{-5}	$a^{5.433}$
H–H anomaly at $\lambda=1/2$	1.04×10^{-4}	$a^{5.464}$
$\lvert\lambda^*-1/2\rvert$	4.5×10^{-5}	$a^{5.535}$

3.3 Regulator dependence

A separate mode-resolved audit showed that the finite-spacing H–H defect tracks the order of the difference stencil. For a fixed smooth mode, second-, fourth-, and sixth-order stencils gave approximately $a^{1.94}$, $a^{3.84}$, and $a^{5.76}$ convergence, respectively. This observation is important for the later quantum analysis: a finite lattice violates the continuum Leibniz rule, so a nonzero regulator-level closure residual should not be identified with a physical quantum anomaly unless it survives refinement.

3.4 Operational records and emergent local SO(3) gauge

The v1.1 revision removes one assumption used in the earlier phase-space construction. Instead of postulating a local internal frame gauge, we begin from an operational record map. At a nondegenerate graph cell the primitive triad e_a^i contains nine real components, whereas the local spatial Gram record $h_{ab}=e_a^i e_b^i$ contains six. Linearizing the map $e\mapsto h$ gives rank six and a three-dimensional kernel for every tested generic triad. The kernel agrees with $\delta e=e\Omega$, $\Omega^T=-\Omega$, to a maximum subspace error 6.83×10^{-16}. At finite amplitude, equality of the metric record within a fixed orientation sector implies $e_2=e_1O$ with $O\in SO(3)$. Thus the standard local frame redundancy can be interpreted as non-uniqueness of the primitive representation of the same metric record, rather than an independently inserted gauge group.

$$h = e\,e^T, \quad \delta h = \delta e\, e^T + e\,\delta e^T, \quad \ker(D\mathcal{R}_e) = \{ e\Omega \mid \Omega^T = -\Omega \}. \tag{3.3}$$

The tangent space also splits directly from the record map. Writing $\delta e=e\Xi$ and $\Xi=S+\Omega$ with $S^T=S$ and $\Omega^T=-\Omega$ gives $\delta h=2eSe^T$, while the Ω component drops out identically. The symmetric part is therefore record-changing (horizontal) and the antisymmetric part record-invisible (vertical). For two symmetric primitive strain generators S_1 and S_2, the commutator $[S_1,S_2]$ is antisymmetric. Numerically, the raw order difference between the two finite strain updates scales as $\varepsilon^{2.005}$ whereas the metric-record difference scales as $\varepsilon^{3.010}$. This primitive scheduler diamond provides a finite-step example in which the leading noncommutativity of two physical record-changing updates lies entirely in the emergent SO(3) fiber.

3.5 Spatial address gauge from relational graph records

The spatial analogue is formulated using relational records rather than coordinate labels. A labeled long-wavelength graph patch may be represented by addresses x^a and metric components $h_{ab}(x)$, but the physical local records are adjacency, proper link lengths, link inner products/angles, and relational matter data. Under a smooth address change $y=F(x)$ with Jacobian $J=dF/dx$, preservation of a complete local set of link Gram records forces the usual tensor transformation law. In a direct reconstruction, three independent two-dimensional link-length records determine the transformed metric and reproduce $J^{-T} h J^{-1}$ with a maximum relative error 6.01×10^{-16}. Local link lengths and angles are invariant at machine precision, while an incorrect control in which the addresses are changed but h is not transformed produces an $O(\varepsilon)$ physical record change.

$$y = F(x), \quad q' = J\,q, \quad h'(y) = J^{-T}\, h(x)\, J^{-1}, \quad q'^T h' q' = q^T h q. \tag{3.4}$$

Infinitesimally, the transformed metric approaches $-\mathcal{L}_V h$ in the passive-coordinate convention with first-order finite-step error. The same redundancy survives in finite weighted graphs: proper edge-length records and the associated weighted-Laplacian spectrum are unchanged at numerical precision under a smooth relabeling when all records are transported consistently. A conceptual distinction is essential. For a finite relational record R on n vertices, the physical state is the isomorphism class $[R]=S_n\cdot R$. Its stabilizer Aut(R) is the exact isometry group of that particular state, not the gauge group. Generic weighted records in the tests have Aut(R)=1 even though their labeled orbit contains n! representatives. This is the finite counterpart of a generic continuum metric having no Killing symmetry while still possessing coordinate/diffeomorphism redundancy.

$$[R] = S_n \cdot R, \quad \mathrm{Aut}(R) = \{ p \in S_n \mid p\cdot R = R \}, \quad |[R]| = n!/|\mathrm{Aut}(R)|. \tag{3.5}$$

To identify the low-energy smooth sector inside the much larger permutation quotient, we considered refinement sequences whose physical displacement field has bounded discrete strain. Exact row-shift permutations approximating the smooth shear $x\mapsto x+A \sin y$ converge to the continuum displacement with RMS error approximately $a^{0.963}$, while random permutations develop a discrete strain approximately $a^{-0.994}$. In independent two-sampling refinement tests, neighbor-derived Jacobians converge as $a^{1.996}$, the reconstructed metric representation as $a^{1.995}$, local Gram records as $a^{1.989}$, global area and scalar-matter integrals as approximately a^2, and the first fifteen nonzero weighted-graph eigenvalues as $a^{2.012}$. These results support the interpretation of continuum spatial Diff(Σ) as the smooth refinement sector of exact finite label/isomorphism redundancy, not as a large finite automorphism group.

3.6 Scheduler non-observability and the normal-refresh algebra

Gauge redundancy alone does not determine the commutator of physical refreshes. A second microscopic consistency condition concerns the internal factorization of one infinitesimal global update. For local unitary factors $U_A=e^{-i\varepsilon A}$ and

U_B=e^−iεB, disjoint support gives exact commutativity, whereas overlapping factors differ first through −ε^2[A,B]. Explicit counterexamples show that unitarity and locality alone do not make the overlap ordering unobservable. The appropriate leading-order condition is instead that the commutator lie in the commutant of the complete physical-record algebra, equivalently that it act only on record-invisible/gauge information (modulo constraints and central phases). In a finite physical⊗gauge model, a pure-gauge commutator changes the full kinematical state at O(ε^2) while the reduced physical state changes only at O(ε^3); the complete physical-observable commutant is reconstructed with the expected dimension and 5.12×10^−16 subspace error.

$$U\_AU\_B - U\_BU\_A = -\varepsilon^2[A,B] + O(\varepsilon^3), \quad [[A,B],O\_phys]=0 \quad \forall O\_phys \Leftrightarrow [A,B]\in\mathcal{A}\_phys'. \tag{3.6}$$

For geometric local-clock refreshes, this criterion becomes directly testable. Two normal deformations δ_NX=Nn and δ_MX=Mn of a genuinely two-dimensional twisted surface have a commutator that is tangential to leading order and is accurately described by the inverse-metric structure function below. At ε=5×10^−5 the normalized commutator error is 6.18×10^−5, the normal component is 5.41×10^−6, and replacing h^{ab} by a fixed background δ^{ab} leaves an O(1) error (1.417). A free coefficient multiplying the structure function fits c=0.999947. After compensating the predicted tangential address shift, the finite scheduler diamond improves from an O(ε^1.997) raw difference to O(ε^2.996), while an invariant area differs only as O(ε^2.992).

$$[\delta\_N,\delta\_M]X = V^a\partial\_aX, \quad V^a = h^{ab}(N\partial\_bM - M\partial\_bN). \tag{3.7}$$

If these local refreshes admit a Hamiltonian realization δ_NF={F,H[N]}, the commutator identity for Hamiltonian vector fields transfers the same structure to the constraints. The crucial logical change relative to the earlier draft is that the normal-normal hypersurface bracket is no longer introduced merely because it is the GR bracket: it follows from normal physical refresh, relational address gauge, and internal scheduler non-observability. The scheduler principle itself remains an explicit microscopic consistency assumption; U4 shows that it cannot be reduced to unitarity and locality alone.

$$\{H[N],H[M]\} = D[h^{ab}(N\partial\_bM - M\partial\_bN)] \quad \text{(mod constraints and finite-regulator terms).} \tag{3.8}$$

3.7 Infrared uniqueness of the ADM/DeWitt Hamiltonian

Once the gauge structure is isolated, the remaining low-energy Hamiltonian freedom is small. Time-reversal invariance and analyticity eliminate odd powers of momentum and make the quadratic term leading. After reduction to a symmetric metric momentum, a direct SO(3) representation-theory audit starts from 21 independent coefficients in a generic symmetric quadratic form and finds invariance rank 19, leaving exactly two quadratic scalars. The recovered invariant subspace agrees with π^{ab}π_ab and π^2 to 6.48×10^−15. At zero and two spatial derivatives, locality, parity, and covariance leave a cosmological term and √h R as the leading configuration scalars. The most general leading Hamiltonian density is therefore the two-parameter kinetic family below, plus higher-order irrelevant operators.

$$\mathcal{H}\_IR = A(\pi^{ab}\pi\_ab - \lambda\pi^2)/\sqrt{h} + \sqrt{h}(c_0 + c\_R\, R) + O(P^4/P\_*^2,\ \ell\_*^2\nabla^4). \tag{3.9}$$

The scheduler-derived hypersurface-deformation structure then selects the DeWitt ratio λ=1/2 in 3+1 dimensions and ties the curvature normalization to the kinetic normalization, leaving the usual gravitational coupling, an optional cosmological term, and boundary terms. This is the same canonical uniqueness structure emphasized in the ADM/DeWitt and geometrodynamics literature [4,5,19]. The GCC-specific contribution is not a new proof of that continuum theorem, but a route by which its premises are reconstructed from the fixed-graph record/scheduler hierarchy and independently recovered in the numerical λ scans.

$$\mathcal{H}\_GR = (\pi^{ab}\pi\_ab - \tfrac{1}{2}\pi^2)/\sqrt{h} - \sqrt{h}\,R + 2\Lambda\sqrt{h} \quad \text{(up to overall gravitational normalization).} \tag{3.10}$$

The statement is an infrared uniqueness result, not exact microscopic uniqueness. P^4 and four-derivative curvature operators are not forbidden; they are suppressed by two additional powers of the corresponding low-energy expansion parameter. This distinction is also important for the quantum theory: the current momentum-quadratic class enjoys the structural Weyl/Moyal cancellation described in Section 7, whereas a genuine P^4 ultraviolet term can reintroduce an ħ^2 correction to the constraint commutator.

3.8 Restricted finite-to-continuum refinement proposition

The strongest finite-to-continuum statement obtained so far is intentionally restricted. Let $h \in C^3(T^2)$ be uniformly positive definite, let $F \in C^3(T^2)$ be orientation-preserving and bi-Lipschitz, and let T_a be a shape-regular periodic triangulation with mesh a. Construct two graph sequences sampling the same continuum geometry, one on T_a and the other on $F(T_a)$, with the transformed metric $h_B(F(x))=DF(x)^{-T} h_A(x) DF(x)^{-1}$. Midpoint Taylor expansion predicts a second-order relative error for straight-edge proper-length records because the even quadratic term in the mapped centered chord cancels. A vertex-based straight-chord simplex area has only first-order relative invariance locally, while centered global geometric and smooth scalar integrals recover second-order convergence.

$$d_{edge}(A_a,B_a) \le C_e a^2, \quad d_{simplex}(A_a,B_a) \le C_T a, \quad d_{global}(A_a,B_a) \le C a^2, \quad \delta h \to -\mathcal{L}_V h. \tag{3.11}$$

The corresponding triangulated audit gives exponents 1.998 for the mean edge record, 0.997 for the local triangle-area record, 1.995 for the global area, and 1.999 for a scalar-matter relational integral. The normalized error coefficients remain bounded throughout the tested refinement family. This is presented as a restricted proposition with a proof sketch based on Taylor expansion, uniform ellipticity, bi-Lipschitz control, and shape-regularity—not as a completed theorem for arbitrary GCC graphs. A formal proof would still require precise record-distance definitions, general triangulation families, chart control, record-separation results, and extension to tensor/spin/gauge matter records.

Box 1. Updated logical route from fixed-graph records to infrared ADM gravity.

Layer	Result in the updated construction
Primitive local record	$h=e\,e^T$ defines six metric records from nine triad components; the three-dimensional record kernel is local SO(3).
Spatial relational quotient	Finite node labels are representational; smooth bounded-strain refinement sequences recover Lie-dragged spatial address redundancy.
Scheduler consistency	Disjoint local factors commute; overlapping order defects must be invisible in the physical record algebra and therefore land in gauge.
Normal-refresh commutator	The gauge shift is $V^a=h^{ab}(N\partial_b M-M\partial_b N)$, yielding the normal-normal hypersurface-deformation structure.
Low-energy operator basis	Time reversal, analyticity, covariance, and two-derivative truncation reduce the leading Hamiltonian to the DeWitt one-parameter kinetic family plus Λ and R.
Infrared result	Constraint closure fixes $\lambda=1/2$, giving ADM/GR as the leading universality class. This is not yet a unique microscopic update theorem.

4 Matter universality and the full classical first-class algebra

4.1 Scalar, Maxwell, and Yang–Mills coupling

To test whether different matter sectors select a common spatial metric rather than sector-dependent propagation geometries, independent coefficients were introduced in the spatial-gradient or magnetic terms. The target value is unity when the matter sector propagates on the same h_{ab} that appears in the gravitational structure function. Scalar, Maxwell, and non-Abelian SU(2) tests all drive these coefficients toward one under refinement.

Table 3. Matter-coupling universality tests. The finite-grid roots approach the common-metric value.

Sector	Continuum selection	Convergence	Representative finest result
Scalar, massless	$\xi^*\to 1$	root error $\sim a^{3.80}$	finest $\xi^*=1.0088$
Scalar, massive/quartic	$\xi^*\to 1$	root error $\sim a^{3.97}$	finest $\xi^*=1.0224$
Maxwell, low mode	$\zeta^*\to 1$	root error $\sim a^{4.00}$	finest $\zeta^*=1.0162$
Maxwell, higher mode	$\zeta^*\to 1$	root error $\sim a^{4.18}$	finest $\zeta^*=1.0486$
SU(2), Gauss-exact non-Abelian	$\zeta^*\to 1$	root error $\sim a^{3.47}$	N=17: $\zeta^*=0.999798$

For the Gauss-exact SU(2) branch the non-Abelian field-strength contribution was about 9.9% at $g_{YM}=1.5$, while the Gauss residual was at machine level. Varying the self-coupling up to $g_{YM}=3$ increased the non-Abelian fraction to roughly 20% without materially shifting the continuum ζ selection. Thus the result is not merely the Maxwell limit in disguise.

4.2 SU(2) first-class constraint algebra

An off-shell color-mixed branch was used to test the internal gauge algebra, while a Gauss-exact branch was used for the physical hypersurface-deformation tests. The expected semidirect structure was recovered, including the Gauss correction in the gauge-covariant spatial-constraint bracket.

$$\{G[\alpha],G[\beta]\} = g\, G[\alpha\times\beta] \tag{4.1}$$

$$\{C[V],C[W]\} = C[[V,W]] - G[\kappa], \quad \kappa^I=V^a W^b F^I\_ab \tag{4.2}$$

$$\{H[N],H[M]\} = D\_cov[h^{ab}(N\partial\_bM-M\partial\_bN)] \tag{4.3}$$

Table 4. Representative SU(2)+gravity classical constraint-algebra audit (N=15).

Bracket	Finest normalized anomaly	Convergence
{G,G}	3.93×10^{-14}	machine precision
{C,G}	2.31×10^{-17}	machine precision
{G,H_total}	5.07×10^{-7}	a^3.72
{D,H_total}	1.21×10^{-4}	a^3.80
{H_total,H_total}	7.90×10^{-3}	a^3.68

4.3 Fermionic geometry

Fermions require a frame rather than only a metric. From h_ab we construct a co-triad e_a^i, a torsion-free spin connection ω_a^{ij}, and a spinor connection Ω_a. The tetrad postulate converges under refinement; the Clifford identity {γ^a,γ^b}=2h^{ab} holds at machine precision; local frame covariance of the Dirac operator is recovered; and spin curvature converges to the curvature of the same emergent geometry. At N=15 the tetrad-postulate residual was about 5×10^{-5}, while the metric-from-triad and Clifford errors were at $\sim10^{-16}$.

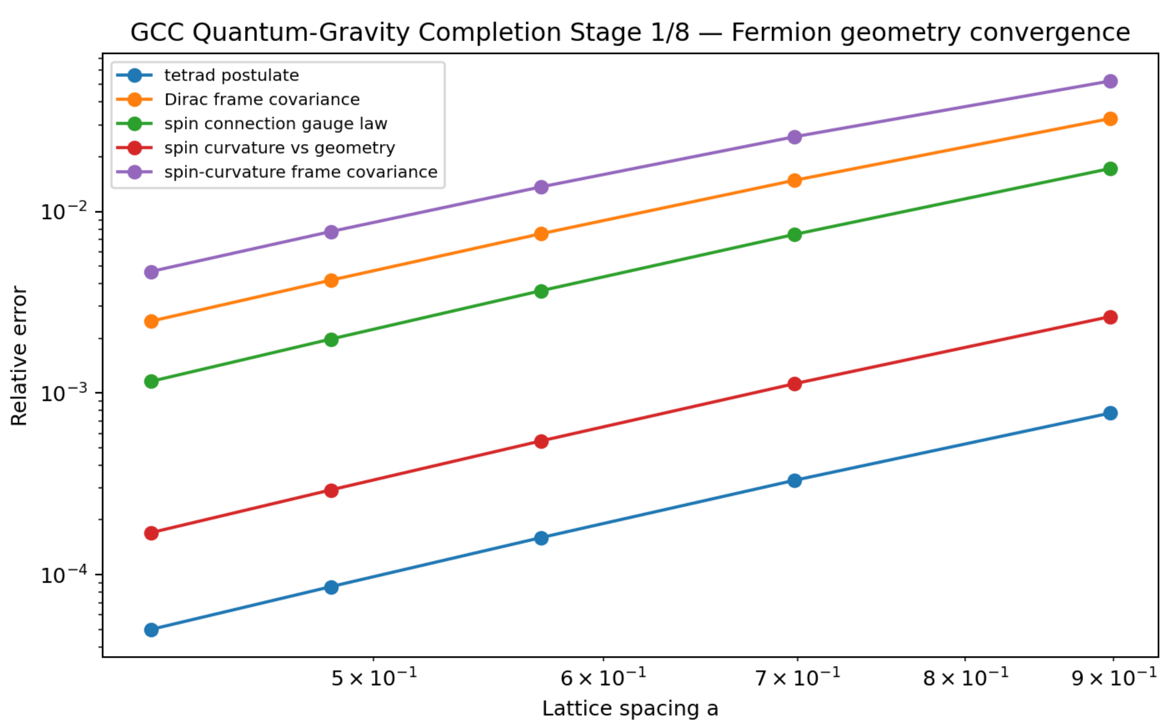

Figure 1. Fermionic geometric audit. Tetrad, local-frame, and spin-curvature diagnostics converge as the fixed graph is refined.

5 Primitive triad phase space and microscopic refresh candidate

5.1 Variables below the metric

The effective metric phase space is lifted to a primitive local-frame phase space with nine triad components e_a^i and nine conjugate momenta P^a_i per cell. The gauge-invariant metric and symmetric momentum are composites,

$$h\_ab = e\_a^i\, e\_b^i, \quad \pi^{ab} = ¼(P^a{}_i\, e^{ib} + P^b{}_i\, e^{ia}) \tag{5.1}$$

with local internal-frame constraint

$$J\_ij = P^a{}_i\, e\_a^j - P^a{}_j\, e\_a^i = 0. \tag{5.2}$$

The primitive phase space has 18 dimensions per cell. Three first-class SO(3) generators remove two dimensions each after constraint plus gauge quotient, leaving 12 dimensions—the same as six h_ab plus six π^ab. On the rotation-constraint surface, P^a_i=2π^{ab}e_b^i and the canonical one-form satisfies P^a_i de_a^i=π^{ab}dh_ab. Numerical reconstruction, Poisson-map, and Hamiltonian-flow audits agreed at approximately 10^{-15}–10^{-19}.

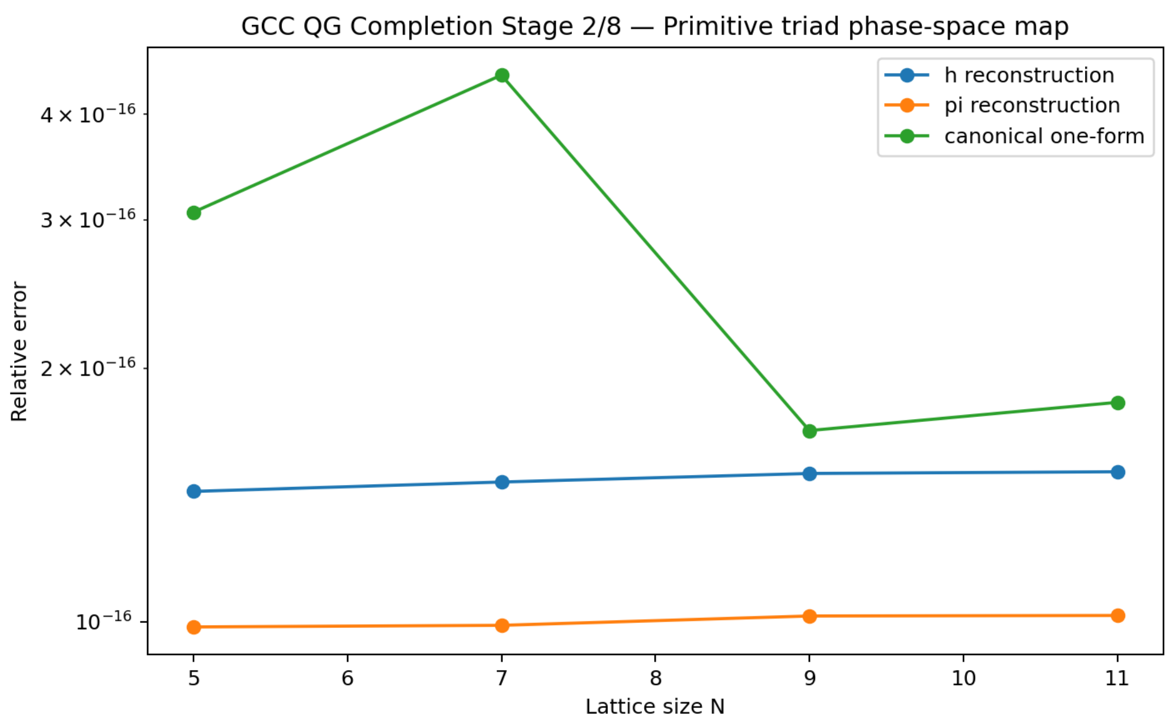

Figure 2. Primitive triad-to-metric phase-space map. Reconstruction and canonical-one-form errors are at numerical precision.

5.2 Low-energy microscopic Hamiltonian and refresh

The low-energy primitive Hamiltonian is defined as the pullback of the previously selected ADM-type Hamiltonian to (e,P):

$$H\_micro[N] = \Sigma\_x\, a^3\, N\_x\, [(\pi^{ab}\pi\_ab - ½\pi^2)/\sqrt{h} - \sqrt{h}\; R\_link[e] + H\_matter]. \tag{5.3}$$

This is a concrete microscopic candidate but not a unique derivation from the original GCC axioms. Its Hamiltonian flow is local with respect to the finite curvature stencil, symplectic, reversible, and invariant under the internal SO(3) frame symmetry. The associated classical Koopman refresh is norm-preserving on phase-space amplitudes; the physical quantum update is introduced only after canonical quantization.

6 Canonical quantization and operator ordering

6.1 Kinematical quantization

On a finite graph the formal Schrödinger representation is

$$[\hat{e}\_a^i(x), \hat{P}^b{}_j(y)] = i\hbar\, \delta\_a^b\, \delta^i{}_j\, \delta\_xy/a^3. \tag{6.1}$$

No finite-dimensional matrix representation can satisfy the ordinary canonical commutator exactly because the trace of a commutator vanishes. Numerical regulators therefore use the exponentiated Weyl algebra, XZ=e^{-2πi/N}ZX, which was satisfied at $\sim 10^{-16}$. The local primitive kinetic Hessian has one negative DeWitt conformal mode, three zero SO(3) gauge modes, and five positive shape modes at every tested site.

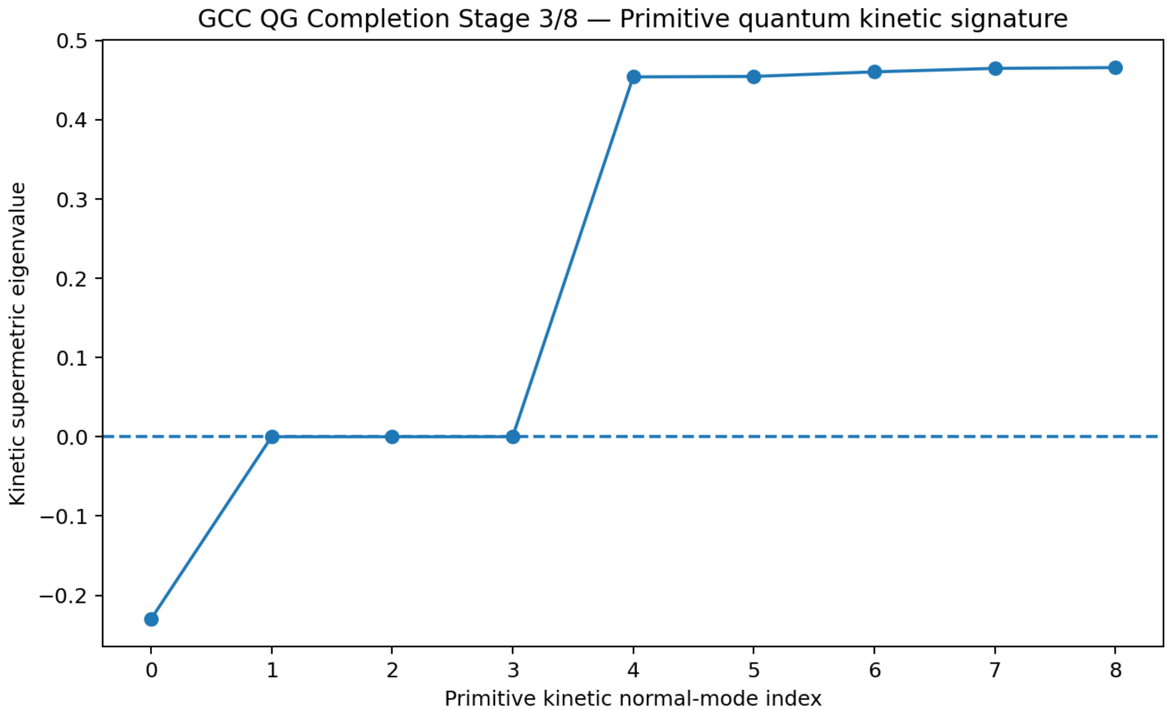

Figure 3. Primitive kinetic supermetric spectrum at a representative full-three-dimensional background site. The signature is one negative conformal direction, three gauge-zero modes, and five positive shape modes.

The negative conformal direction does not by itself imply non-unitary evolution: self-adjointness, not positivity of the Hamiltonian spectrum, controls unitarity. A finite spectral regulator gave one-tick unitary errors below 10^{-14} across the normal modes.

6.2 Weyl ordering as the primary closure prescription

A divergence-form ordering $\hat{P}\ A(\hat{e})\ \hat{P}$ is manifestly Hermitian and was numerically unitary, but its Weyl symbol differs from the intended classical symbol by an $\hbar^2$ configuration-space term. Because the central question is constraint closure, the primary operator prescription in this paper is Weyl quantization, Op_W(C). This makes the relation between operator commutators and Moyal brackets explicit.

7 Quantum constraint closure

7.1 Structural Moyal cancellation

For Weyl symbols, the operator commutator is represented by the Moyal bracket,

$$[\hat{H}\_N,\hat{H}\_M]/(i\hbar) \leftrightarrow \{H\_N,H\_M\} - (\hbar^2/24)\Lambda^3(H\_N,H\_M) + O(\hbar^4). \tag{7.1}$$

The current GCC low-energy Hamiltonian has the form H[N]=Σ_x N_x[K_x(e_x,P_x)+U_x(e-neighborhood)], where each K_x is local and at most quadratic in the momenta, and U is momentum-independent. For this class all higher odd Moyal terms in the H–H commutator vanish. U–U contains no momentum derivatives; K_x–K_y is disjoint for x≠y; the same-cell K–K star commutator vanishes; and any K–U contribution of Moyal order three or higher would require at least three momentum derivatives of K, which is impossible for a quadratic kinetic term.

The same argument applies to the tested internal and spatial generators, which are quadratic in the primitive canonical variables for external smearings. Hence the Weyl quantum-symbol algebra inherits the classical finite-spacing closure defect but receives no additional intrinsic ħ-dependent anomaly in the current momentum-degree≤2 class.

7.2 Direct matrix test and regulator interpretation

A genuinely noncommuting two-cell finite-torus Hamiltonian was constructed and $[H_N,H_M]/(i\hbar)$ was compared with the Weyl-quantized classical bracket. The full operator norm is contaminated by the finite spectral wrap/Nyquist boundary, but after projection onto a low-momentum sector that cannot reach the wrap boundary the relative mismatch falls to $\sim 10^{-14}$. In the full three-dimensional inherited closure test, the sixth-order regulator gives a finest H–H symbol anomaly of 8.30×10^{-4} and convergence approximately a^5.46.

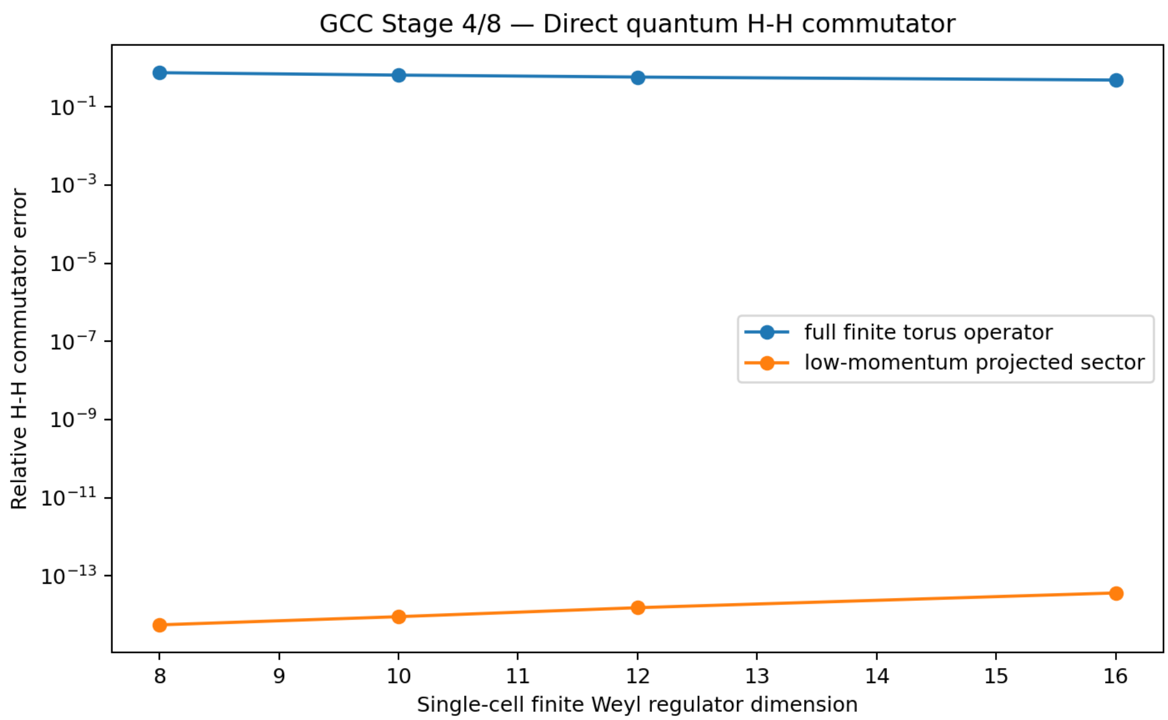

Figure 4. Direct two-cell quantum H–H commutator audit. The finite-torus full norm contains a spectral-edge artifact; the low-momentum projected sector agrees with the Weyl-quantized classical bracket at numerical precision.

A control with an ηP^4 ultraviolet term produces a nonzero Λ^3 and therefore a genuine leading $\hbar^2$ correction. This control is critical: the anomaly cancellation is not generic but is a structural property of the selected quadratic-momentum completion. The ultraviolet rule chosen below is consequently designed not to introduce P^4 terms.

8 Physical Hilbert space and the universal clock

8.1 Positive master constraint

The physical state space is defined by a positive master constraint, following the general strategy of combining multiple first-class constraints into a single positive operator [10,11]:

$$\hat{M} = \Sigma\_x\, a^3[\hat{J}\dagger W\_J\hat{J} + \hat{G}\dagger W\_G\hat{G} + \hat{D}\dagger W\_D\hat{D} + \hat{H}\_\perp\dagger W\_H\hat{H}\_\perp], \qquad \mathcal{H}\_\text{phys}=\ker \hat{M}. \tag{8.1}$$

In the linearized graviton sector the kernel is explicit. For each nonzero lattice momentum q_i=2sin(k_i a/2)/a, the transverse-traceless projector has rank two. Across ten generic momentum directions the positive polarization master constraint had nullity two, the kernel projector agreed with the analytic TT projector to $\sim 10^{-15}$, and projected polarizations were transverse and traceless at numerical precision.

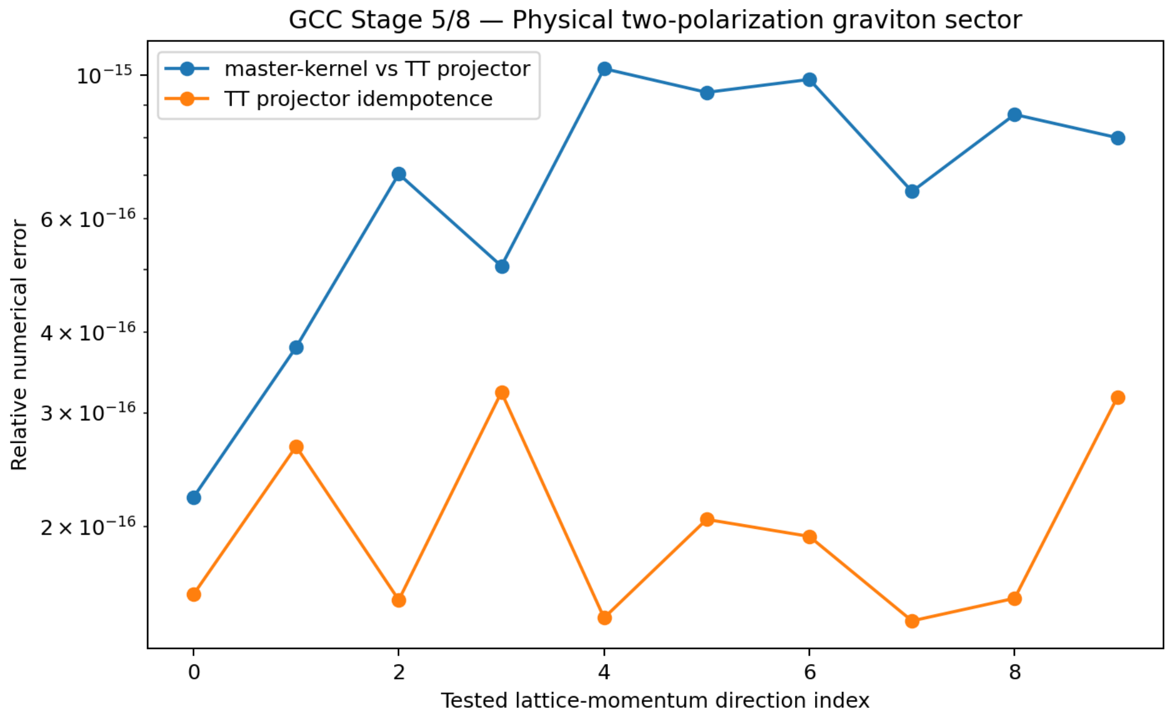

Figure 5. Physical one-graviton sector. The positive master-constraint kernel coincides numerically with the rank-two transverse-traceless projector.

8.2 Universal-K history states

A direct equation $\hat{H}|\Psi\rangle=0$ together with the same $\hat{H}$ as the generator of K evolution would create a frozen-state ambiguity. The present completion therefore treats the universal update index as a relational/discrete history variable, in the spirit of conditional-time constructions [12]. A physical history is

$$|\Psi\_hist\rangle = \Sigma\_K\, |K\rangle\otimes|\psi\_K\rangle, \quad |\psi\_\{K+1\}\rangle = \hat{U}\_GCC\, |\psi\_K\rangle. \tag{8.2}$$

The update relation is imposed as a propagation constraint on clock$\otimes$system Hilbert space. In a 12-step, four-dimensional physical system, the propagation constraint had nullity four—exactly the underlying system dimension—and an explicit history state had residual 6.65×10^{-17}. A combined spatial-plus-update master constraint retained exactly four physical states for two momenta times two TT polarizations. The regulated projector $\exp(-\tau\hat{M})$ converged toward the physical projector.

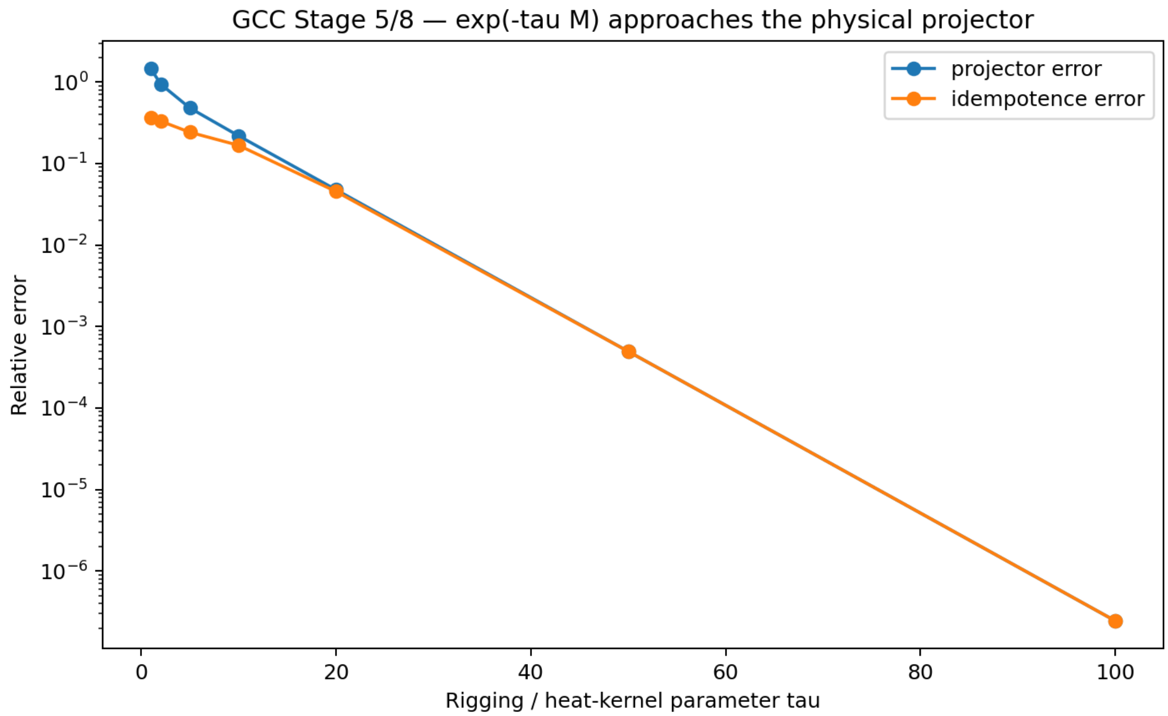

Figure 6. Heat-kernel/rigging-map regulator $\exp(-\tau\hat{M})$ approaching the physical projector in the finite history-state model.

9 Semiclassical general relativity and low-energy quantum field theory

9.1 Graviton coherent states

Each physical linearized TT mode is a harmonic oscillator. Coherent states therefore provide an explicit test that the physical quantum state space possesses a classical gravitational-wave limit. For a mode frequency ω, the expectation values obey the classical oscillator equations,

$$\langle Q(t)\rangle = Q_0 \cos\omega t + (P_0/\omega)\sin\omega t,\quad \langle P(t)\rangle = P_0 \cos\omega t - \omega Q_0 \sin\omega t. \tag{9.1}$$

In the regulated calculation the maximum Ehrenfest error was 2.13×10^{-14}. Relative coherent-state fluctuations scale exactly as N̄^{-1/2}, so large occupation yields a classical tensor field. This provides an explicit physical-state semiclassical check in addition to the earlier classical ADM correspondence.

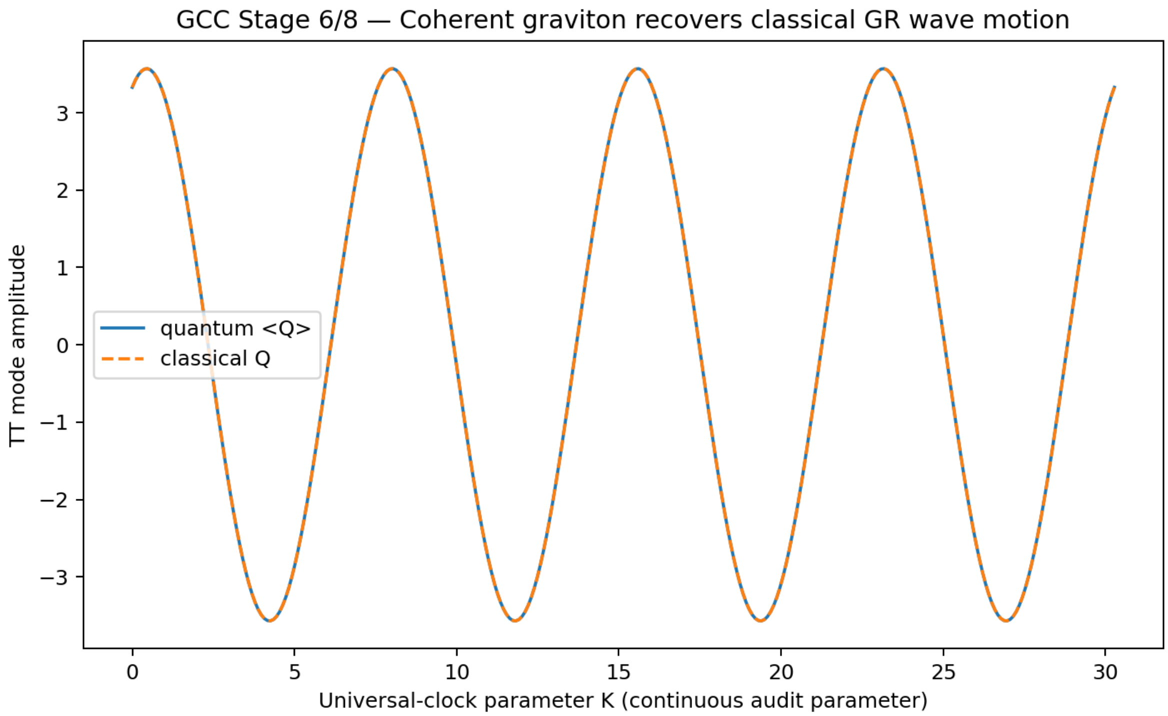

Figure 7. Expectation value of a coherent physical TT graviton mode compared with the classical gravitational-wave trajectory.

9.2 QFT dispersion and common causal cone

The fourth-order centered derivative used in the full-three-dimensional audits has Fourier symbol $q_i^{(4)}=[8\sin(k_i a)-\sin(2k_i a)]/(6a)=k_i+O(a^4)$. Scalar, Maxwell, linearized Yang–Mills, naive Dirac, and TT-graviton modes therefore recover the usual relativistic dispersions in the continuum. Measured dispersion errors were approximately $a^{3.996}$ for the massless bosonic sectors and naive Dirac field. A Wilson term removes the standard lattice fermion doubler but reduces the tested finite-a convergence to approximately $a^{1.99}$, a conventional lattice-fermion trade-off rather than a GCC prediction [6,7].

$$\omega^2 = m^2 + h^{ab}\, p_a\, p_b \tag{9.2}$$

For fermions the curved-space principal symbol satisfies $(\gamma^a p_a)^2=h^{ab}p_a p_b I$ with maximum numerical error 3.38×10^{-16} over generic positive-definite metrics. The massless group-speed error converges approximately as $a^{3.995}$, so the tested graviton, scalar, Maxwell, Yang–Mills, and fermion sectors share the same low-energy causal cone. Bosonic low-occupation commutators and fermionic anticommutators recover the standard Fock-mode algebras.

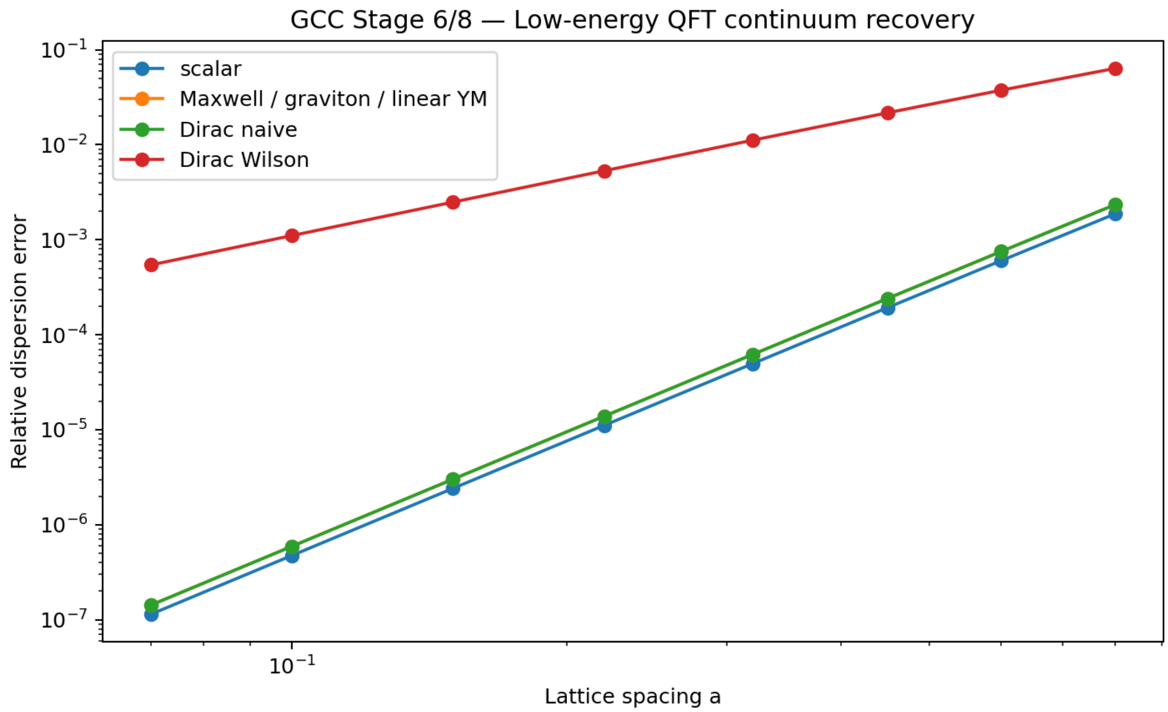

Figure 8. Low-energy QFT dispersion recovery. The common fourth-order derivative gives $O(a^4)$ continuum convergence for the naive low-energy sectors; the Wilson fermion regulator trades this for doubler removal.

10 Bounded-load ultraviolet completion and regular black holes

10.1 Saturation rule without higher momentum powers

The ultraviolet completion is chosen to preserve the momentum-degree≤2 structure that protected the Stage-7 quantum constraint algebra. Instead of adding P^4 terms, physical configurations are restricted to a bounded invariant load domain $0\leq\chi\leq1$ with a self-adjoint no-flux boundary at saturation. A finite-domain quantum model remained Hermitian and unitary, with a 250-tick norm drift of 3.64×10^{-14}. A nonlinear configuration-only saturation control retained $\Lambda^3(H\_N,H\_M)=0$.

The variable χ should be interpreted as a coarse-grained invariant load rather than a new material field fixed by the original GCC papers. Its precise microscopic definition is an open derivation problem. The present v1 branch supplies one explicit realization that is consistent with a universal finite-response scale.

10.2 Spherical regular-core realization

The explicit static spherical completion adopts the Paper-III/Hayward-type mass response [3,13] with a core scale tied to a universal saturation length ℓ_* by $r\_c^3=2M\ell\_*^2$:

$$m(r)= M\, r^3/(r^3+2M\ell\_*^2),\quad f(r)=1-2m(r)/r. \tag{10.1}$$

Near the center $f(r)=1-r^2/\ell\_*^2+O(r^5)$, so the core is de Sitter-like. The exact central curvature and density are

$$R(0)=12/\ell\_*^2,\quad R\_{abcd}R^{abcd}(0)=24/\ell\_*^4,\quad \rho\_*=3/(8\pi\ell\_*^2)\ (G=c=1). \tag{10.2}$$

The normalized density/load $\chi=\rho/\rho\_*$ remains between zero and one. The effective anisotropic stress tensor is covariantly conserved. The radial null-energy condition is saturated, the tangential null-energy condition is non-negative, and the strong-energy combination changes sign in the inner core. Thus the classical energy-condition assumption entering standard singularity theorems is relaxed without introducing a central curvature divergence.

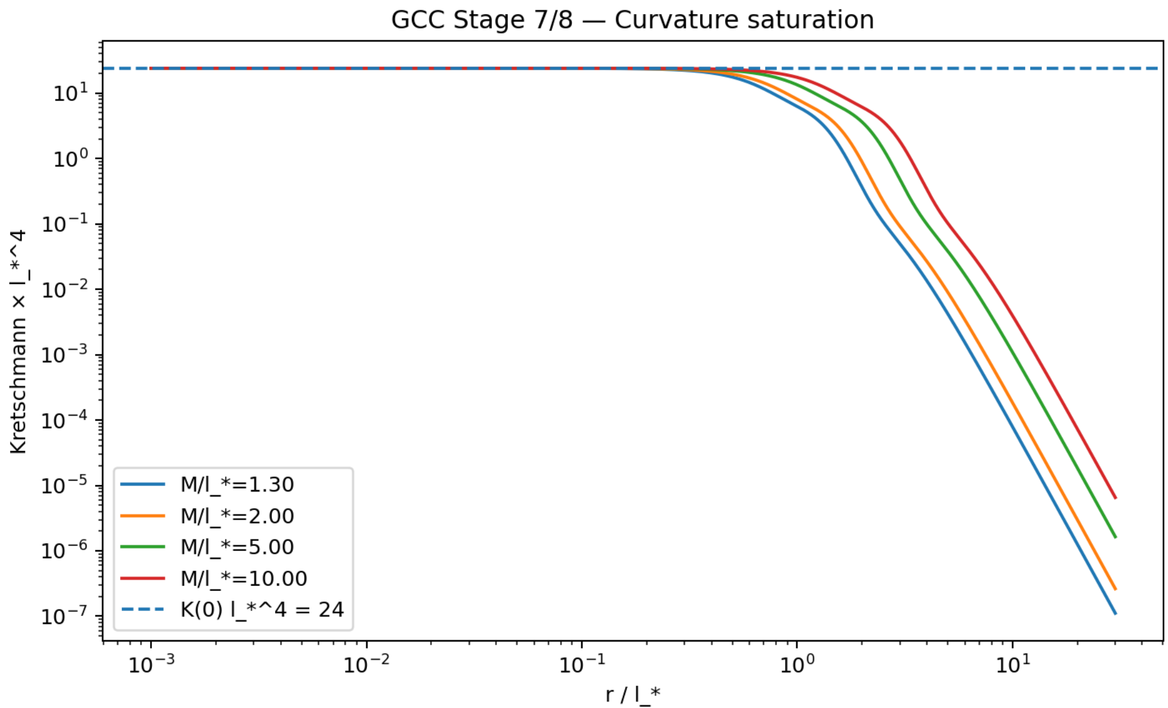

Figure 9. Kretschmann curvature for the selected spherical regular-core branch. The scanned family saturates at the universal central value $K\ell_*^4=24$.

10.3 Horizons, temperature, and information accounting

$$r_h^3 - 2Mr_h^2 + 2M\ell_*^2 = 0. \tag{10.3}$$

The extremal double root occurs at $r_{crit}=\sqrt{3}\ell_*$ and $M_{crit}=(3\sqrt{3}/4)\ell_*$ in geometric units. The Hawking temperature inferred from surface gravity is

$$T_H(r_h) = (r_h^2-3\ell_*^2)/(4\pi\, r_h^3), \tag{10.4}$$

which vanishes at the extremal endpoint and reaches a maximum at $r_h=3\ell_*$, $M=27\ell_*/16$. If $\ell_*=\ell_P$, the endpoint area entropy is $S_{ext}=3\pi$. A large initial black hole has entropy far exceeding this value, so a finite remnant cannot by itself store the entire initial information budget. Combined with the unitary history-state construction, the selected branch therefore requires information transfer into radiation/history correlations during evaporation. No microscopic Page curve is derived here.

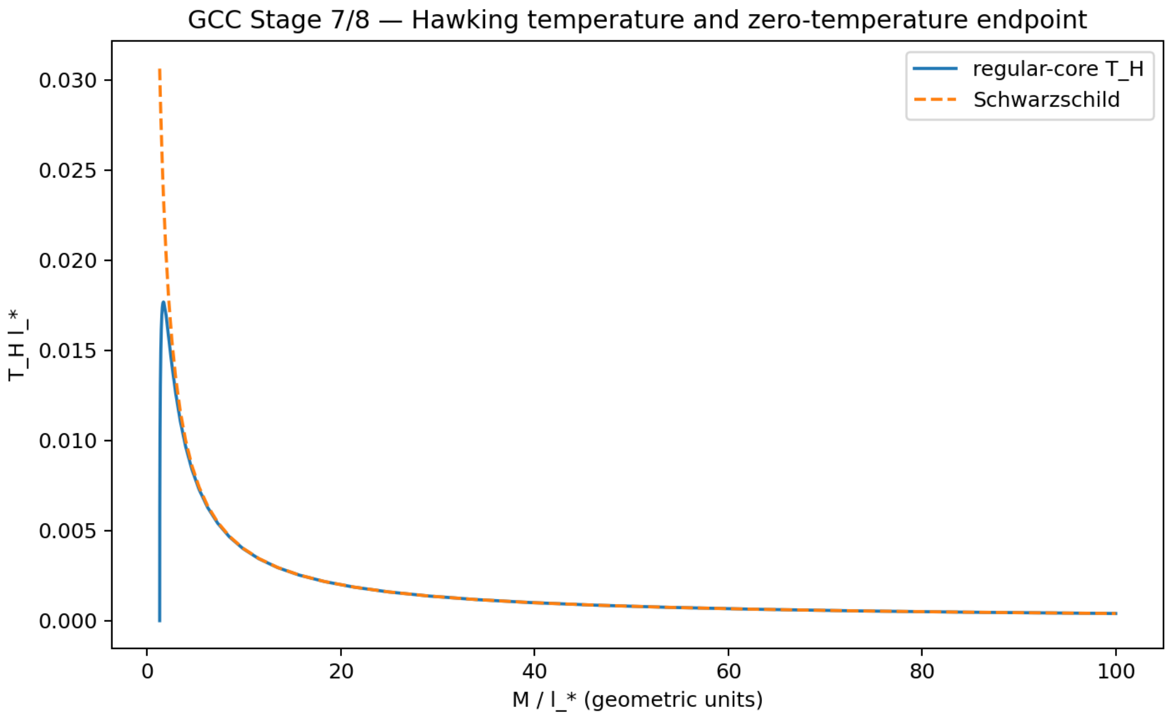

Figure 10. Hawking-temperature curve of the selected regular-core branch. The Schwarzschild limit is recovered at large mass while the temperature turns over and approaches zero at the extremal endpoint.

11 Falsifiable predictions of the selected v1 strong-gravity branch

11.1 Leading exterior corrections

At large radius the static metric is

$$f(r)=1-2M/r + 4M^2\ell_*^2/r^4 + O(r^{-7}). \tag{11.1}$$

Defining $\varepsilon=(\ell_*/M)^2$, the leading one-scale shifts are

Table 5. Leading static predictions of the selected one-scale regular-core branch.

Observable	Leading v1 result	Fractional shift
Outer horizon	$r_+/M = 2 - \varepsilon/2$	$\Delta r_+/r_{GR} = -\varepsilon/4$
Photon sphere	$r_{ph}/M = 3 - 4\varepsilon/9$	$\Delta r_{ph}/r_{GR} = -4\varepsilon/27$
Shadow impact parameter	$b_{ph}/M = 3\sqrt{3}[1-2\varepsilon/27]$	$\Delta b/b_{GR} = -2\varepsilon/27$
Eikonal real frequency	$M\Omega_{ph} = (3\sqrt{3})^{-1}[1+2\varepsilon/27]$	$\Delta\Omega/\Omega_{GR} = +2\varepsilon/27$
Lyapunov/damping proxy	$M\lambda_{ph} = (3\sqrt{3})^{-1}[1-4\varepsilon/27]$	$\Delta\lambda/\lambda_{GR} = -4\varepsilon/27$
ISCO radius	$r_{ISCO}/M = 6 - 11\varepsilon/18$	$\Delta r/r_{GR} = -11\varepsilon/108$
ISCO frequency	$M\Omega_{ISCO} = (6\sqrt{6})^{-1}[1+29\varepsilon/216]$	$\Delta\Omega/\Omega_{GR} = +29\varepsilon/216$

The common scaling $\propto M^{-2}$ is itself a falsifiable pattern for universal ℓ_*. Near the extremal endpoint the perturbative expansion is no longer valid; solving the exact static geometry gives order-percent effects. At $M=M_{crit}$ the static branch gives approximately a −5.37% shadow shift, +5.68% eikonal real-frequency shift, −15.35% Lyapunov-scale shift, and +9.87% ISCO-frequency shift relative to Schwarzschild.

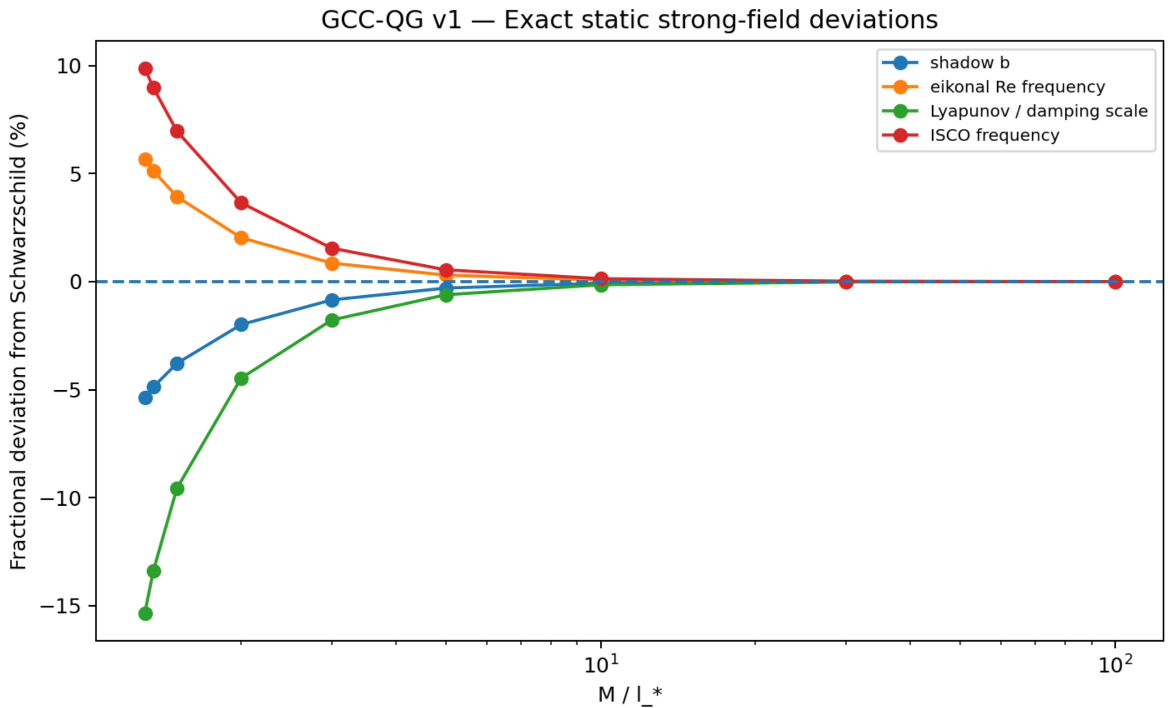

Figure 11. Exact static deviations from Schwarzschild across the regular black-hole branch. Effects become appreciable only when M approaches a few times the ultraviolet length scale in geometric units.

11.2 Present observational context

The Event Horizon Telescope reports that the Sgr A* image size is within about ten percent of Kerr predictions [16]. Current gravitational-wave spectroscopy is likewise consistent with general relativity: the GW250114 public analysis reports a fundamental ringdown frequency measured to about two-percent precision and a decay time to about nine percent, both consistent with Kerr/GR [18], while the GWTC-4.0 remnant tests report no strong evidence for deviations and no evidence for post-merger echoes in the analyzed events [17]. These data do not presently provide a rigorous bound on the v1 parameter because the selected GCC strong-gravity solution is static and spherical whereas astrophysical black holes rotate.

For the minimal identification $\ell_*=\ell_P$, the exterior deviations for stellar and supermassive black holes are fantastically small because $\varepsilon=(m_P/M_{phys})^2$. A 30-solar-mass black hole has a static ringdown correction of order 10^{-80}, and supermassive systems are even more suppressed. Thus present astrophysical agreement with GR is expected rather than surprising. The distinctive v1 signatures become order-percent only for masses within a few Planck masses if $\ell_*=\ell_P$.

The selected regular-event-horizon branch does not make gravitational-wave echoes mandatory. Echoes would require additional reflecting or horizonless structure and should therefore not be presented as a robust prediction of GCC-QG v1. This is a correction to earlier exploratory echo-oriented possibilities.

11.3 Falsification conditions

Table 6. Examples of direct theoretical or observational falsification conditions for the selected v1 branch.

Test	Clean falsifier
Quantum closure	A non-vanishing continuum anomaly for the stated Weyl-ordered momentum-degree≤2 constraint class.
UV domain	Failure to construct a self-adjoint nonlinear bounded-load domain compatible with the constraints.
Critical endpoint	Verified evaporation/horizon physics continuing below M_{crit} with no extremal endpoint for the same universal ℓ_*.
Exterior scaling	A controlled low-spin non-GR deviation with mass scaling incompatible with $(\ell_*/M)^2$ for the one-scale branch.
Sign pattern	A clean near-static measurement showing shadow enlargement, lower orbital/ringdown frequency, or larger ISCO radius with the opposite correlated pattern.

Test	Clean falsifier
Curvature/load cap	A realization of the selected branch exceeding its universal local saturation bound while remaining inside the defined physical domain.

12 Discussion: what has been derived, selected, and left open

12.1 What is strongest in the present construction

The strongest part of the program is now the long-wavelength consistency chain rather than any single black-hole ansatz. A primitive triad carries an operational metric record whose kernel reconstructs local SO(3); a finite labeled graph is quotiented by record-preserving isomorphism, while bounded-strain refinement sequences provide a smooth spatial address sector approaching Lie dragging; local-factorization consistency requires scheduler defects to lie in record-invisible directions; and normal geometric refreshes then generate the inverse-metric hypersurface structure function. Low-energy analyticity and covariance reduce the kinetic freedom to the DeWitt family, while closure selects $\lambda=1/2$. Independently, centered norm-preserving transport reconstructs the Levi-Civita connection, link curvature feeds the same ADM-type Hamiltonian, multiple matter sectors select one spatial metric, the primitive triad phase space maps canonically to the effective metric phase space, and the current Weyl-ordered momentum-quadratic quantum Hamiltonian introduces no intrinsic higher-Moyal H–H anomaly. The agreement of these independent routes is stronger than simply inserting ADM gravity as a starting assumption.

The physical-state construction is less complete but mathematically definite: the master constraint defines a physical kernel, a non-empty perturbative TT sector is explicit, and universal-clock history states provide a consistent way to combine constrained global histories with conditional unitary evolution. The semiclassical limit is similarly explicit in the perturbative sector: coherent gravitons recover classical wave motion and the tested matter sectors recover relativistic QFT mode structure on the common metric.

12.2 What remains completion-dependent

The remaining completion-dependence is now sharply localized. The exact microscopic refresh law is not unique; internal scheduler non-observability is a physically motivated consistency principle rather than a theorem of the original GCC papers; and the finite-to-continuum spatial-gauge argument has reached only a restricted proposition with a proof sketch, not a general graph-refinement theorem. The ultraviolet black-hole sector is likewise an effective completion rather than a derivation. The scaling r_c^3=2Mℓ_*^2 follows from imposing a universal central density/curvature scale on the chosen Hayward-type profile, but the rational radial profile itself is an ansatz. A dynamical collapse calculation has not shown that the microscopic GCC update uniquely produces this form. Likewise, the equality ℓ_*=ℓ_P is the minimal identification suggested by the Planck-calibrated graph but remains an assumption until the saturation scale is derived microscopically.

Rotation is the most important phenomenological extension. Real EHT and LIGO/Virgo/KAGRA sources are rotating, so a Kerr-like regular GCC solution and its perturbation spectrum are needed before the static shadow and eikonal formulas can be used for precision parameter estimation. Inner-horizon stability, nonlinear collapse, evaporation, and a microscopic Page curve also remain open. These are substantial research problems, but they are extensions of a specified candidate rather than undefined placeholders in the v1 architecture.

12.3 Relation to other quantum-gravity approaches

GCC-QG v1.1 is not intended as a replacement-by-analogy for loop quantum gravity, causal sets, Regge calculus, lattice gauge theory, quantum cellular automata, or other discrete approaches. It shares ingredients with several of them: combinatorial structure, link transport, canonical constraints, finite Weyl regulators, relational-time ideas, and refinement questions [6–12,20]. Its distinctive organization is the universal grid-clock substrate, the reconstruction of gauge as redundancy of operational graph records, the scheduler route to the hypersurface structure, the explicit numerical selection of the GR constraint structure, and the decision to preserve a momentum-quadratic ultraviolet kinetic class while imposing saturation through the physical configuration domain.

13 Conclusion

This paper has assembled an explicit quantum-gravity candidate extension of Grid-Clock Cosmology and, in the v1.1 revision, has strengthened the origin of its classical gauge sector. The construction starts from a fixed graph and universal update ordering; interprets local SO(3) and spatial address freedom as redundancies of operational records and labeled graph representations; uses scheduler non-observability to route normal-refresh commutators into the emergent spatial gauge; and reduces the leading covariant, analytic Hamiltonian to the ADM/DeWitt universality class. It then reconstructs curvature, demonstrates low-energy matter universality, introduces primitive triad variables, quantizes them with a Weyl prescription, controls the current quantum constraint algebra, defines a physical Hilbert space, recovers semiclassical GR and relativistic QFT in the tested sector, and selects a bounded-load ultraviolet branch with finite black-hole curvature and falsifiable strong-field signatures. The result is not a proof that quantum gravity is solved or that GCC is realized in nature. It is a concrete fixed-graph candidate whose remaining assumptions—especially microscopic scheduler non-observability, general refinement-gauge convergence, ultraviolet profile selection, rotation, and full nonlinear spectral control—are explicit enough to be attacked or falsified in subsequent work.

The appropriate conclusion is therefore not that quantum gravity has been solved, but that GCC now possesses a concrete, internally organized, and testable canonical quantum-gravity candidate architecture. Its most important next tests are a rotating strong-gravity completion, a dynamical microscopic derivation of saturation, a full nonlinear analysis of the physical master constraint, and a controlled evaporation calculation. Failure of the quantum constraint algebra, the bounded-load construction, or the correlated one-scale strong-field predictions would directly falsify the selected v1 completion.

Appendix A. Eight-stage quantum-gravity completion audit

Table 7. Eight-stage completion roadmap summarized with the scientific scope of each completion claim.

Stage	Gate	v1 status	Key output
1/8	Fermion / tetrad / spin connection	Complete at geometric gate	Dirac/frame covariance and spin curvature converge; Clifford algebra at machine precision.
2/8	Primitive triad refresh	Complete as low-energy candidate	Canonical $(e,P)\rightarrow(h,\pi)$ map and dynamic projection at numerical precision.
3/8	Canonical quantization	Complete as finite regulator	Weyl algebra, Hermitian normal modes, unitary update.
4/8	Quantum constraint closure	Complete for current Weyl P^2 class	No intrinsic higher-Moyal anomaly; finite-grid anomaly converges.
5/8	Physical Hilbert space	Complete at construction/perturbative level	Master constraint, TT kernel, history-state clock.
6/8	Semiclassical GR + QFT	Complete perturbatively	Coherent gravitons, common causal cone, QFT mode algebras.
7/8	UV / black-hole saturation	Complete as selected effective branch	Bounded load, finite core, remnant endpoint, anomaly-safe P^2 UV rule.
8/8	Predictions / falsification	Complete	Static one-scale predictions, current-context audit, explicit falsifiers.

Appendix B. Provenance and assumptions

Table 8. Provenance of the principal ingredients. This distinction is essential for avoiding overstatement.

Item	Scientific status
Fixed graph + universal K	Original GCC architecture [1]
Levi-Civita / ADM / DeWitt / TT / Weyl / master-constraint mathematics	Established mathematics/physics [4,5,10,11]
Information-norm and no-spurious-translation interpretation	Exploratory GCC reconstruction principle
Primitive triad e,P completion	Exploratory completion developed in this program
History-state interpretation of K	Exploratory completion choice, related conceptually to relational-time constructions [12]
Bounded load $0\leq\chi\leq1$	New v1 ultraviolet completion axiom
$m(r)=Mr^3/(r^3+2M\ell_*^2)$	Selected effective Hayward-type branch [3,13]
$\ell_*=\ell_P$	Minimal optional identification; not uniquely derived

Item	Scientific status
Operational triad record quotient $h=e\,e^T$ → local SO(3)	Standard quotient geometry applied as a GCC operational-record interpretation; numerically audited in v1.1
Finite label quotient and smooth refinement sector	Exploratory GCC relational-gauge construction; finite/permutation and refinement tests U6–U9
Internal scheduler non-observability / local factorization principle	New microscopic consistency principle in v1.1; not implied by unitarity and locality alone
Restricted refinement proposition	Proof sketch supported numerically; not a general theorem for arbitrary GCC graphs

Appendix C. Numerical reproducibility summary

All numerical results reported here were generated in explicit finite regulators and retained as CSV/figure artifacts during the completion program. Convergence claims are based on grid or refinement sequences rather than a single finite lattice. The principal diagnostics include derived-curvature and DeWitt-root scans, scalar/Maxwell/SU(2) universality tests, full SU(2) constraint brackets, fermionic spin-geometry convergence, primitive phase-space maps, Weyl-regulated normal modes, direct quantum H–H matrix checks, physical TT/master-constraint tests, semiclassical and QFT dispersion audits, regular-core curvature/thermodynamic scans, and the U1–U9 foundational extension set. The latter includes SO(3)-invariant kinetic-basis reconstruction, normal-refresh commutator tests, scheduler diamonds, physical-observable commutant reconstruction, triad record kernels, relational link invariance, finite graph orbit/stabilizer tests, smooth permutation/refinement convergence, and the restricted triangulated refinement proposition. These artifacts are numerical evidence for the stated construction and convergence behavior; they do not replace analytic proofs where the paper explicitly labels a result as a proposition or proof sketch.

Appendix D. Foundational gauge and infrared-uniqueness extension audit

The v1.1 revision consolidates nine focused extensions that were performed after the initial quantum-completion draft. Their role is to reduce the number of continuum GR structures assumed at the start of the construction. Table D1 records the strongest conclusion that can presently be drawn from each extension. The sequence should be read as a chain of increasingly microscopic reinterpretations, not as nine independent proofs of general covariance.

Table D1. Foundational extensions U1–U9 incorporated into GCC-QG v1.1.

Extension	Question	Principal result	Status
U1	Why ADM at low energy?	SO(3)-invariant quadratic kinetic space is two-dimensional; closure selects the DeWitt combination.	IR uniqueness statement; not microscopic uniqueness.
U2	Can the H–H structure function emerge?	Two local normal refreshes generate $V^a=h^{ab}(N\partial_b M-M\partial_b N)$; $c\to 1$ numerically.	Geometric/path-independence reconstruction.
U3	Why should refresh order be gauge?	Scheduler diamond closes after tangential compensation: $O(\varepsilon^2)$ raw difference → $O(\varepsilon^3)$ physical difference.	Local Factorization Independence formulation.
U4	Does locality imply scheduler independence?	No for overlaps; physical independence requires [A,B] to lie in the physical-observable commutant.	Positive criterion plus explicit counterexample.
U5	Can internal gauge be defined operationally?	$\ker(e\mapsto ee^T)=e\cdot so(3)$; finite equal-record triads differ by SO(3).	Local SO(3) reconstructed from records.
U6	Can spatial gauge be defined by records?	Complete link Gram records force $h'=J^{-T} h J^{-1}$ and infinitesimally $\delta h=-\mathcal{L}_V h$.	Long-wavelength spatial address redundancy.
U7	What is the finite graph counterpart of Diff?	Gauge is the label/isomorphism orbit, not Aut(record); generic weighted records have Aut=1.	Conceptual correction; finite quotient clarified.
U8	Does the smooth sector converge?	Two smooth samplings converge in local/global/spectral records at controlled rates; rough random permutations separate.	Constructive finite-to-continuum convergence.
U9	Can the convergence be stated mathematically?	Restricted T^2/bi-Lipschitz/shape-regular proposition: edge $O(a^2)$, simplex $O(a)$, global $O(a^2)$.	Proof sketch, not general GCC theorem.

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